

C03 - AT3 - Tree Traversal challenges

DATE: 14/08/20

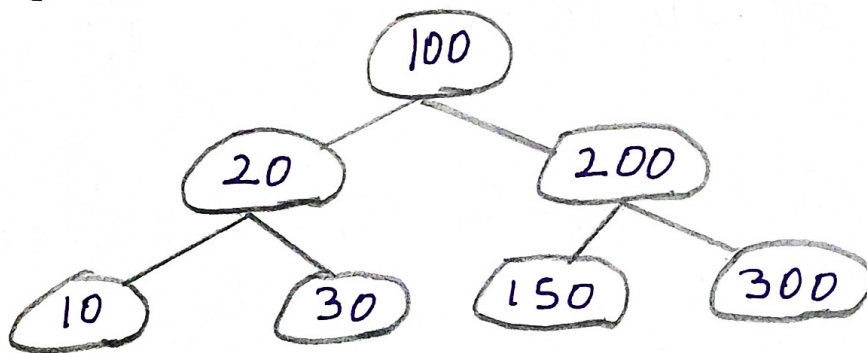
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COURSE: Data Structure

COURSE CODE: CSA0311

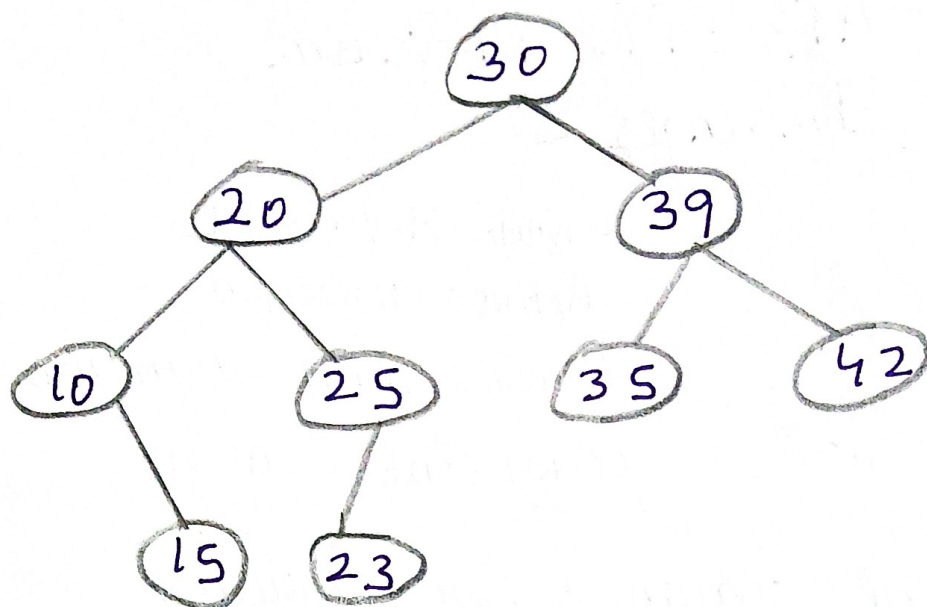
Write the traversal sequences for following trees.



Pre order: 100, 20, 10, 30, 200, 150, 300
[NLR]

Inorder: 10, 20, 30, 100, 150, 200, 300
[LNR]

Post order: 10, 30, 20, 150, 300, 200, 100
[LRN]

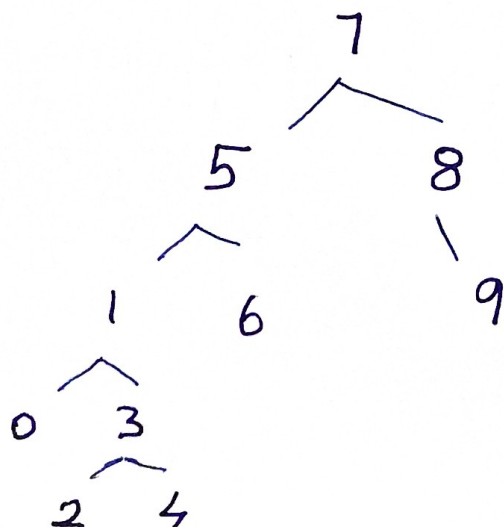


Pre order : 30, 20, 10, 15, 25, 23, 39, 35, 42
[NLR]

Inorder : 10, 15, 20, 23, 25, 30, 35, 39, 42
[LNR]

Post order : 15, 10, 23, 25, 20, 35, 42, 39, 30
[LRN]

Digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is empty. What is the resulting tree's traversal sequence?

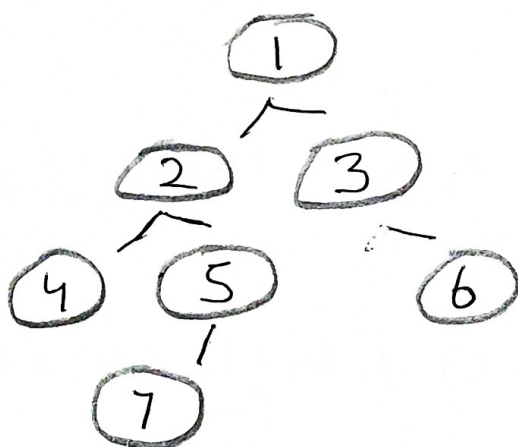


Pre order: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9
[NLR]

Inorder: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
[LNR]

Post order: 0, 2, 4, 3, 1, 6, 5, 9, 8, 7
[LRN]

Question NO: 5

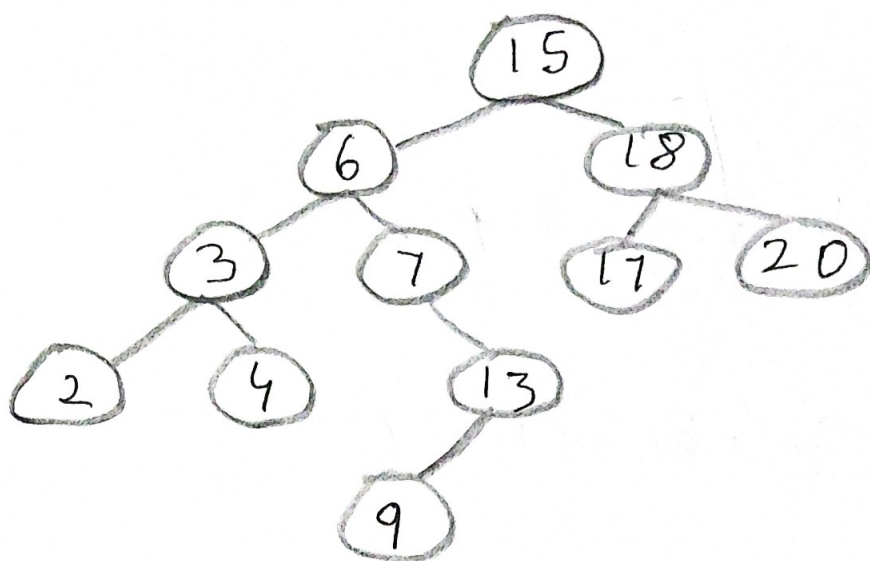


Inorder: 7, 4, 2, 5, 1, 3, 6
[LNR]

Pre order: ~~1, 2, 4, 7, 5, 3, 6~~
[NLR]

Post order: 7, 4, 5, 2, 6, 3, 1
[LRN]

Question no 6:

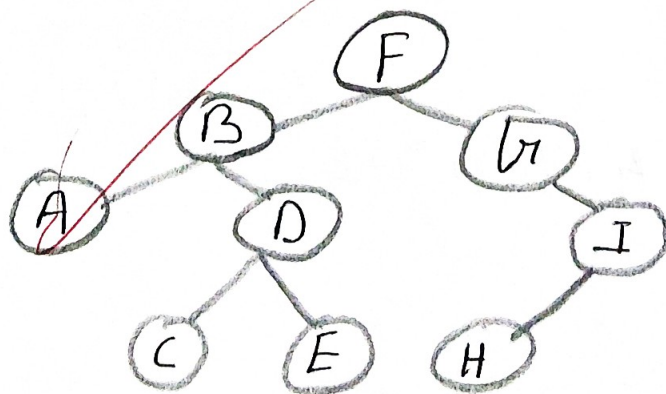


Pre order: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20
[NLR]

In order: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20
[LNR]

Post order: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15
[LRN]

Question 7:



Applying in order traversal on binary tree

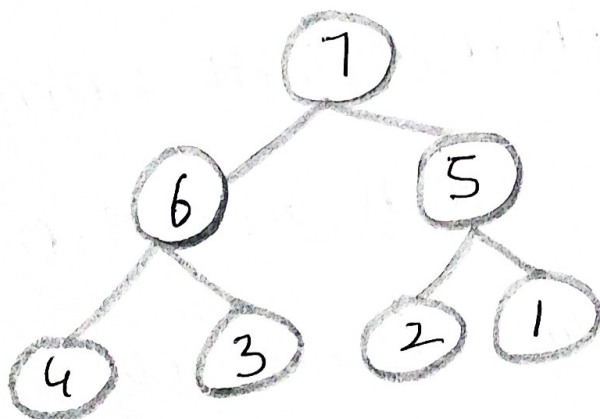
Answer: 3) A, B, C, D, E, F, G, H, I

Inorder - D B E F A G H C

Preorder - A B D E F C G H

What is the Post order?

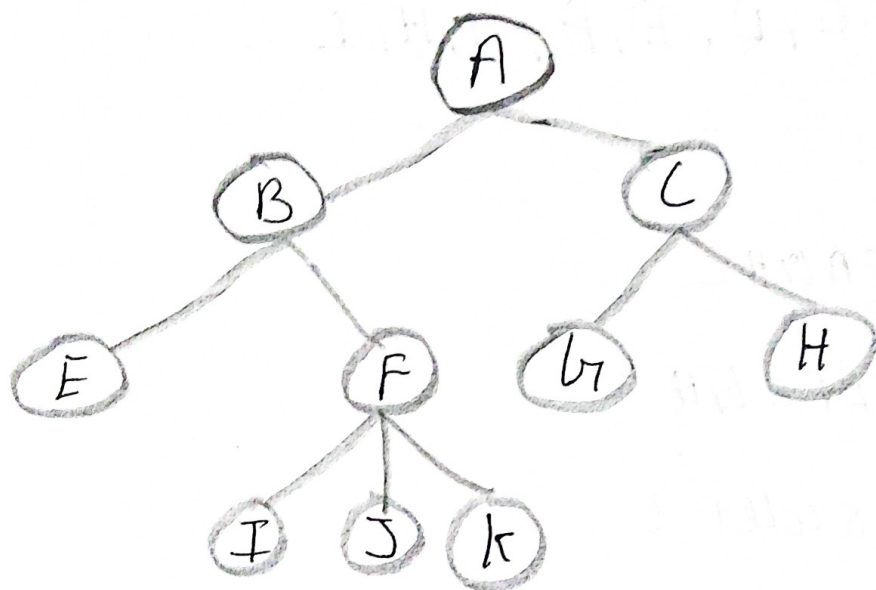
Option 3 - D F E B H G C A



Inorder: 4, 6, 3, 7, 2, 5, 1

Preorder: 7, 6, 4, 3, 5, 2, 1

Postorder: 4, 3, 6, 2, 1, 5, 7



Pre order: A, B, E, F, I, J, K, C, G, H, D

Post order: E, I, J, K, F, B, G, H, C, D, A

In order: E, B, I, F, J, K, A, G, C, H, D

∴ In order is normally defined only for binary tree. In this question is multinary tree, there is no standard traversal.